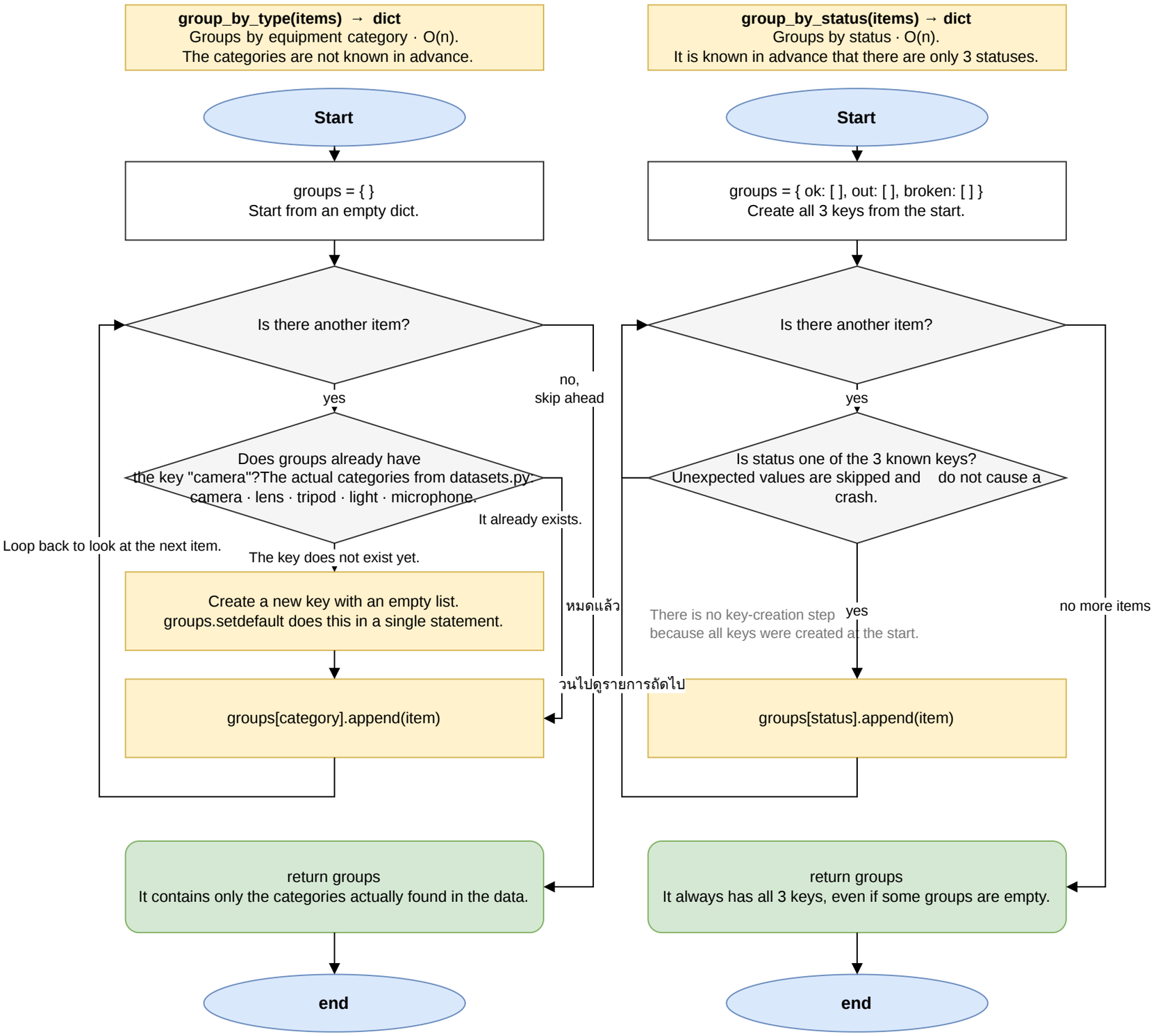
Flowchart 7 — Grouping: group_by_type and

ort_group.py · Receives items as a parameter and does not import anyone else's files on the team',It goes through the data in a single pass like count_by_status, but stores the whole record in a bucket instead of just counting.



Why a single pass with bucketing is faster than searching one category at a time.

• The bucketing method reads the data in one pass no matter how many categories there are, so it is O(n).
•Searching one category at a time (scanning for only "camera" first, then scanning again for "lens") has to re-read the whole dataset once per category, so it is O(n × number of categories).
• The team's dataset has 5 categories (camera · lens · tripod · light · microphone).So searching one category at a time must read the data 5 times to get the same result.
• Both functions use O(n) memory because every item is stored in one bucket or another. No item is discarded, unlike counting, which uses O(1).**)**
• Unlike Neo's count_by_status, which stores only the count, this one stores the whole records. It is used when you need to know "what is there," not just "how many."
• Both functions always return the keys they promise. — group_by_status group_by_status returns all 3 keys even when some groups are empty,
so that anyone using it later does not have to write .get() guards everywhere. Because someone would surely forget one somewhere.